Understanding Data Augmentation Versus Dataset Size in Convolutional Neural Networks Through Internal Representation Analysis

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Introduction

Convolutional Neural Networks (CNNs) typically require large labelled datasets for strong performance. When training datasets are limited, performance degradation is often attributed solely to the insufficient sample size. Augmentation is a method used to mitigate the decline in performance by transforming existing images to reinforce relevant visual patterns, yet effectiveness is often only measured through final accuracy, leaving its impact on the internal learning process of the model often poorly understood.

In this work, we investigate differences in model perception between augmenting smaller datasets versus enlarging the dataset's quantity. Through a representational perspective, the work examines the benefits of each solution and the potential advantages of focusing on smaller dataset quality compared to that of larger unoptimized datasets. Using ResNet 18, we compare models trained with less data with and without augmentation to models trained on larger, unaugmented datasets. This work visually represents global average pooling (avgpool) layer activations using dimensionality reduction techniques as well as silhouette scores to study interclass separation and intraclass consistency. This approach emphasizes the role of representation stability in low-data learning and highlights the importance of internal analysis for understanding model optimization in low-data settings.

ResNet 18 Architecture

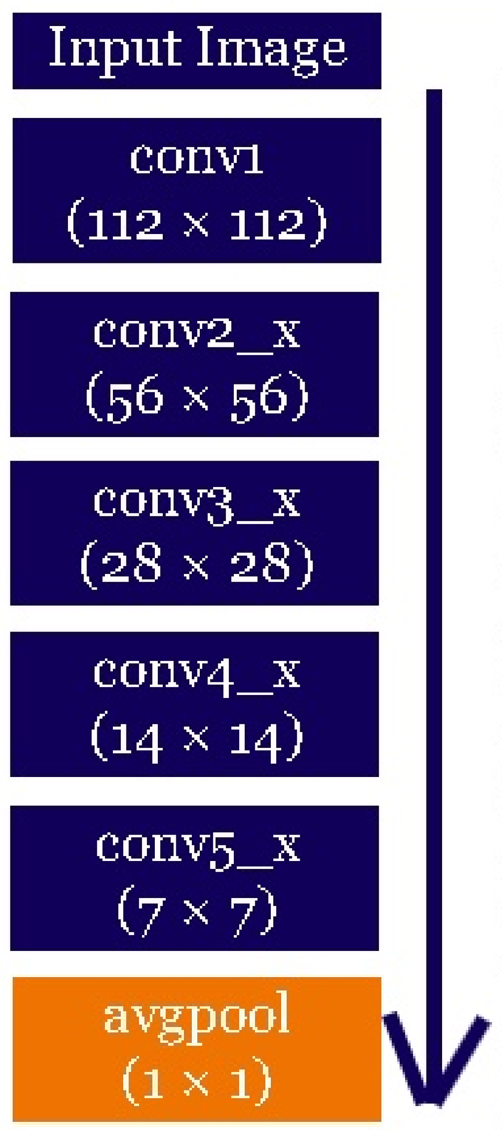


Figure 1 (Left):
Architecture for ImageNet adapted from original ResNet paper [1] . Resnet 18 consists of an initial convolutional layer followed by four stages of residual blocks with skip connections and a final fully connected layer for a total of 18 layers. Building blocks and their formats in brackets with the numbers of blocks stacked displayed in the graph. We pull activations from the avgpool layer and flatten it, highlighted orange to the left.

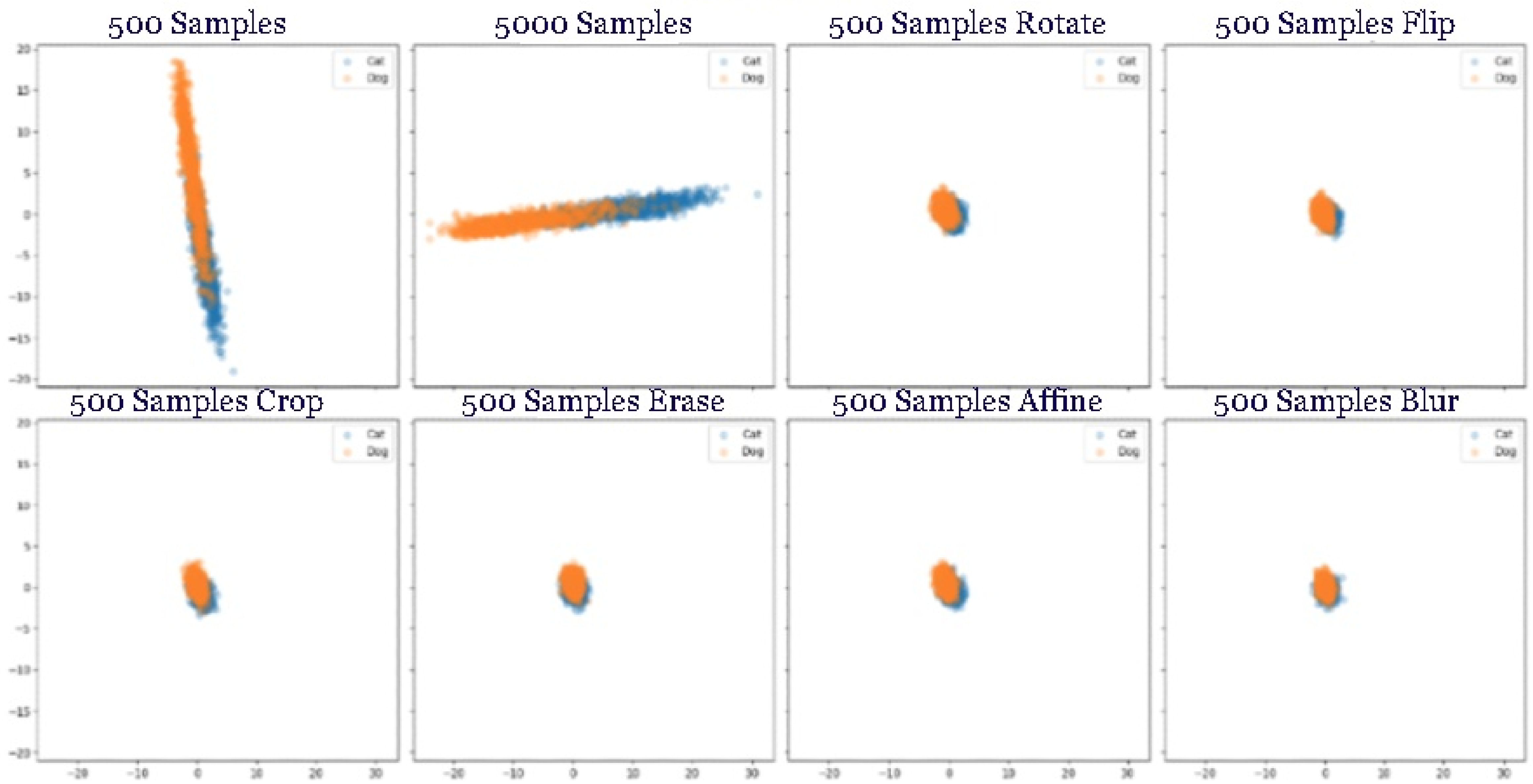
Methodology

Pretraining:
All models used ResNet 18 models initialized with weights pretrained on ImageNet for general purpose visual features before task specific training. I filtered the CIFAR10 dataset to only cats and dogs for a binary subset. Then the subset was used for creating training various training regimes: 1 high image quantity trained model (5000) and 7 low image quantity models (500). 1 high image quantity model and 1 low image quantity model remained unaugmented, the other 6 low image quantity models had transformations as followed: RandomRotation, RandomHorizontalFlip, RandomResizedCrop, RandomErasing, RandomAffine, and GaussianBlur. Treat the unaugmented low image quantity trained model as control.

Supervised Fine Tuning:
The models were fine tuned to distinguish between dog and cat. To reduce overfitting, early layers of the models were frozen while higher layers including layer3, layer4, and the final fully connected layer were fine tuned. Models were trained for 10 epochs using the Adam optimizer with a learning rate of 1e-4 and used the cross entropy loss function.

Representation Extraction and Analysis:
After training, internal representations were extracted from the avgpool layer of the ResNet 18 model using forward hooks during inference. This produced 512-dimensional feature vectors, which were flattened and projected into two dimensions using principal component analysis (PCA) for visualization through matplotlib.

Results

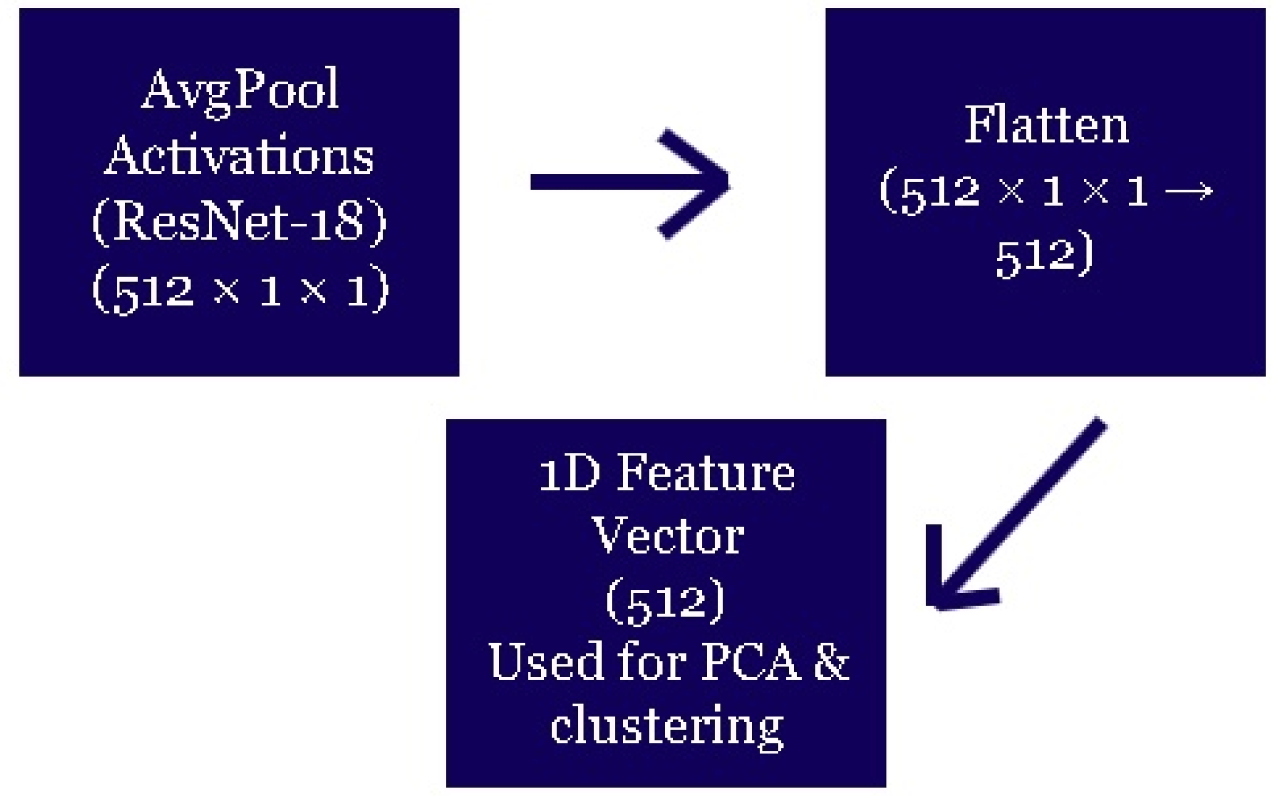


	Silhouette Score	Accuracy (%)	Average Loss
500 Base	0.11108	83.35	0.4842
5000 Base	0.23798	88.45	0.3994
500 Rotate	0.10315	81.00	0.5228
500 Flip	0.12487	84.75	0.4380
500 Crop	0.12493	84.05	0.5011
500 Erase	0.10988	81.70	0.5483
500 Affine	0.11771	82.65	0.5797
500 Blur	0.12044	83.90	0.4805

Figure 1 (Above):
PCA Graph of extracted avgpool activations, showing the internal representation structure learned under different training conditions.

Figure 2 (Left):
Performance and representation metrics across training conditions. The table reports silhouette score, classification accuracy, and average loss for models trained under different data regimes and augmentation strategies.

Activation Flattening Pipeline



Conclusion + Future Work

When examining the average accuracy score of the augmented models, compared to the control model's accuracy of 83.35%, an accuracy of 83.01% represents a small decrease in accuracy. This contrasts with the significantly higher accuracy achieved by the 5000 image model (88.45%). However, this analysis of accuracy alone does not capture the primary effects of augmentation. While training with larger datasets primarily improved interclass separation and accuracy, augmentation consistently resulted in a significant reduction in intraclass variance, indicating more stable internal representations. This reduction in intraclass variance is reflected in the tighter activation clusters of the PCA visualizations and reveals that augmentation can improve model performance not merely by increasing data diversity, but encouraging consistency in how exactly similar inputs are internally encoded. These results display the value of internal analysis for understanding and improving model optimization under data-constrained settings.
Future research could be potentially exploring different augmentation strengths, class counts, and image topics to generalize or expand these results. Evaluating other architectures or freezing strategies can help show how representation stability would depend on model design.

References

[1] He, Kaiming, et al. "Deep residual learning for image recognition." 2016 IEEE Conference on Computer Vision and Pattern Recognition (CVPR), June 2016, pp. 770–778, https://doi.org/10.1109/cvpr.2016.90.